

Phase 1 deliverable (before full RNA-seq processing):

- A short pre-analysis summary (slides or memo/write-up are both fine) covering:

1. Predicted biology if beta-catenin is degraded during stimulation.

Predicted biology:

Background

Beta-catenin is the primary effector for the Wnt signaling pathway. If Wnt signaling is active, β -catenin accumulates in the cytoplasm and moves to the nucleus where it interacts with tcf/lef transcription factors to activate target gene transcription in proliferation and differentiation.

<https://www.nature.com/articles/s41392-025-02142-w>

MYC/CCND1 are typical targets

B-catenin pathway plays important role in differentiation, immune regulation and T-cell development <https://pubmed.ncbi.nlm.nih.gov/31616443/>

Predicted effects of degradation

- Reduced Wnt target gene expression-MYC and ccnd1 transcription effects should be decreased
- Reduced stem-like/memory T-cell populations- Wnt signaling helps maintain populations
- Reduce expression of genes involved in proliferation and growth

Summary: b-catenin degradation will suppress wnt-driven transcriptional populations/alter t-cell differentiation and proliferation

2. Expected acyclovir-only effects (independent of degrader).

Acyclovir is incorporated into DNA and interferes with DNA polymerase.

Transcriptional signatures include:

- DNA damage response pathways
- Nucleotide metabolism pathways
- Cellular stress response signaling

3.Planned comparison matrix

Comparison	Question

Degrader+acyclovir versus untransduced+acyclovir	What changes occur when B-catenin is degraded
Untransduced with/without acyclovir	Effects produced?
2-day versus 5-day activation	What transcriptional changes occur during T-cell activation over periods of time

4. How 2-day vs 5-day activation might differ biologically.

Wnt signaling pathways regulate immune cell development, transcriptional differences between 2 and 5 day activation may reflect immune activation dynamics instead of degrader activity.

5. Proposed additional samples for next round to better separate time, dose, and degrader effects.

Each variable in the next experiment should be tested independently. Some factors are currently partially confounded, making it hard to differentiate between the causes of the degradation, drug treatment, or differences in activation time.

Improved design should include:

- Degrader status: degrader construct present vs untransduced
- Acyclovir dose: no drug, low dose, high dose
- Time: Early activation and later activation (2 vs 5 days)

6. A short literature scan of beta-catenin-in-T-cells and acyclovir-related effects that you think is most relevant to this dataset, including caveats.

B-Catenin

- B-Catenin is the central transcriptional mediator of the Wnt signaling pathway, where it interacts with the tcf/lef transcription factors to regulate gene expression
- Wnt/B-catenin plays an important role in T-cell development, differentiation, and immune responses, pathway promotes memory t-cell formation and maintains stem-like t-cell populations
- B-catenin regulates genes relating to cell proliferation and differentiation, altering its activity may change transcriptional programs.
- Degradation of B-catenin reduces expression of the Wnt target genes and shifts T cells into more differentiated forms.

Acyclovir

Antiviral Drug that inhibits viral DNA polymerase and interferes with viral DNA replication

Transcriptional changes possible independent of B-catenin degradation are related to

- DNA replication/repair pathways
- Nucleotide metabolism
- Stress-response signaling

Caveats included:

- May be difficult to differentiate between degrader specific effects and drug-related effects (produce similar effects, overlap)
- Time/dose partially confounded: may reflect normal t-cell activation rather than degrader activity
- Indirect changes/effects: Some observed genetic/gene expression changes may be downstream effects of t cell differentiation and not directly from Wnt pathway targets

7. An outline of your proposed steps for going from raw RNA-seq FASTQ files to analysis outputs, and what those outputs should include.

Step 1: Quality Control

- Analyze data using Fastqcr or ShortRead in R Studio
- OR use FastQC and then import data into R Studio

Step 2: Trimming- removes unwanted or low-quality parts of sequences reads before analysis

- Recommended using tools outside of R studio such as Cutadapt or TrimGalore

Step 3: Alignment to reference genome

- Use splice-aware RNA sequencer such as STAR or HISAT2

- Produces BAM alignment file

Step 4: Gene Counting

- Tools in R studio such as Rsubread (featureCounts) or GenomicAlignments will work
- Produces gene expression count matrix

Step 5: Differential Expression

- Identifies genes that show significant changes in expression between experimental conditions
- Use tools such as DESeq2 (most common) or edgeR

Step 6: Functional and Pathway Analysis

- Helps to interpret the biological meaning of the differentially expressed genes
- Use tools such as Gene Ontology (GO) enrichment, KEGG pathway analysis, Gene Set Enrichment Analysis (GSEA)

Step 7: Visualization/Reporting

- Various tools can be used to assess different statistics, significance, and summarizations
- Principal Component Analysis (PCA) used to assess sample clustering and overall variation
- Volcano plots showing significantly up- and down-regulated genes
- Heatmaps displaying expression patterns of key genes across samples
- Pathway enrichment plots to summarize the biological pathways affected by β -catenin degradation